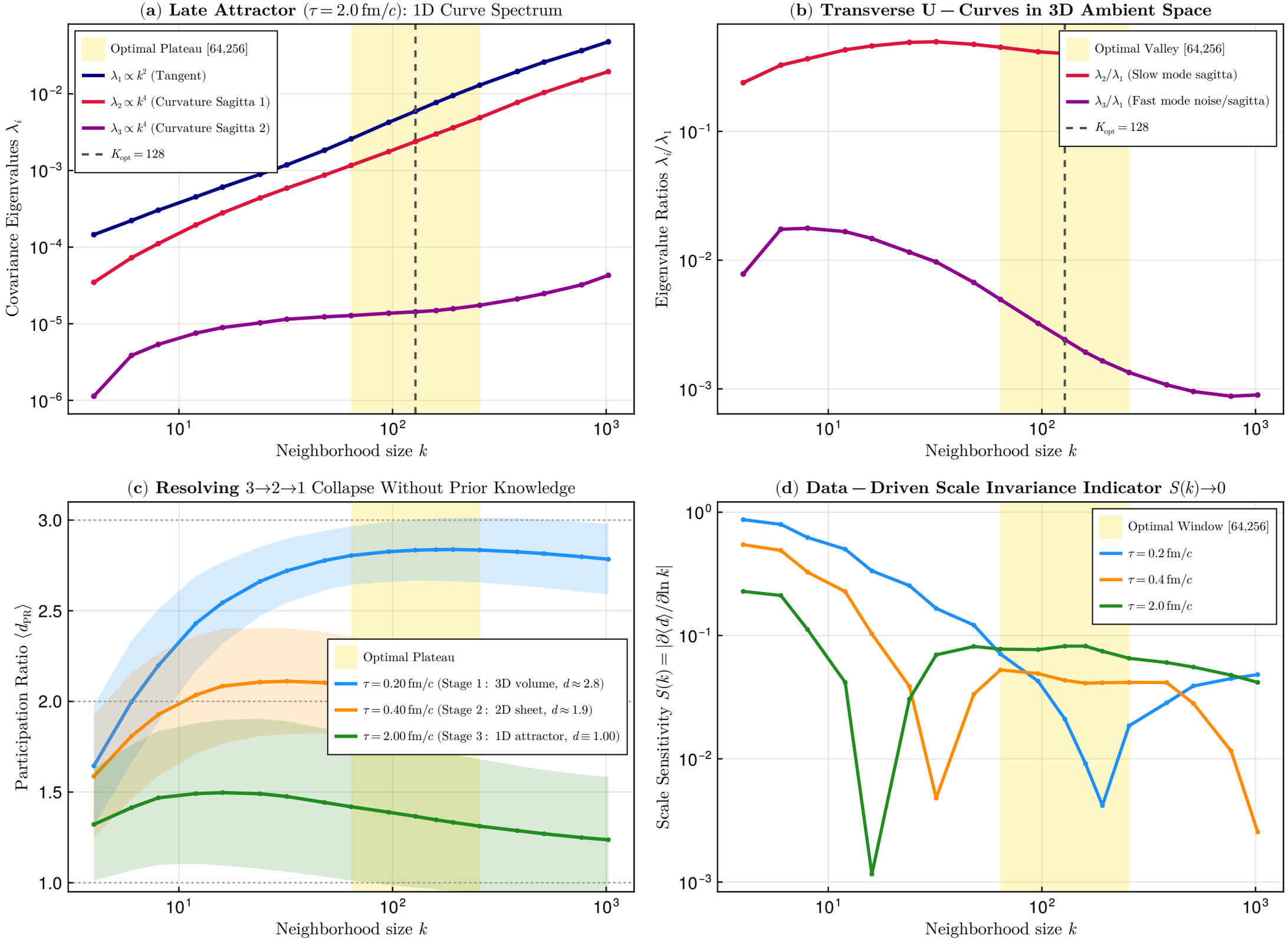
(a) Late Attractor ($\tau = 2.0\,\mathrm{fm}/c$): 1D Curve Spectrum
Optimal Plateau [64,256]
$\lambda_1 \propto k^2$ (Tangent)
$\lambda_2 \propto k^4$ (Curvature Sagitta 1)
$\lambda_3 \propto k^4$ (Curvature Sagitta 2)
$K_{\mathrm{opt}} = 128$
Covariance Eigenvalues λ_i
Neighborhood size k
(b) Transverse U — Curves in 3D Ambient Space
Optimal Valley [64,256]
λ_2/λ_1 (Slow mode sagitta)
λ_3/λ_1 (Fast mode noise/sagitta)
$K_{\mathrm{opt}} = 128$
Eigenvalue Ratios λ_i/λ_1
Neighborhood size k
(c) Resolving 3→2→1 Collapse Without Prior Knowledge
Optimal Plateau
$\tau = 0.20\,\mathrm{fm}/c$ (Stage 1 : 3D volume, $d \approx 2.8$)
$\tau = 0.40\,\mathrm{fm}/c$ (Stage 2 : 2D sheet, $d \approx 1.9$)
$\tau = 2.00\,\mathrm{fm}/c$ (Stage 3 : 1D attractor, $d \equiv 1.00$)
Participation Ratio $\langle d_{\mathrm{PR}} \rangle$
Neighborhood size k
(d) Data — Driven Scale Invariance Indicator $S(k) \to 0$
Optimal Window [64,256]
$\tau = 0.2\,\mathrm{fm}/c$
$\tau = 0.4\,\mathrm{fm}/c$
$\tau = 2.0\,\mathrm{fm}/c$
Scale Sensitivity $S(k) = |\partial\langle d \rangle / \partial \ln k|$
Neighborhood size k